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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
First Semester B.Tech (CBCS) Degree Examinations

December 2024

CY1005-1 – CHEMISTRY OF ENERGY STORAGE AND DISPLAY DEVICES
(For.ACT, EC, EE, VLSI)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

- Which of the following is suitable for powering portable devices such as smartphones and laptops?
A) Lead-acid battery
B) Lithium-ion battery
C) Methanol-oxygen fuel cell
D) Vanadium flow battery
- Which electrolyte is commonly used in lithium-ion batteries?
A) Lithium hexafluorophosphate (LiPF_6)
B) Sodium hydroxide (NaOH)
C) Sulfuric acid (H_2SO_4)
D) Potassium chloride (KCl)
- Photovoltaic cells convert energy to.....energy.
A) Chemical; Electrical
B) Electrical; Chemical
C) Solar; Electrical
D) Electrical; Solar
- A simple molecule having two or more bonding sites through which each can link to other to form a polymer chain is called.....
A) Polymer
B) Monomer
C) Functionality
D) Initiator
- Vinyl compounds are.....
A) Bifunctional monomers
B) Trifunctional monomers
C) Polymers
D) Not example for monomers
- Carbon fibre is obtained from heating
A) Polyvinylchloride
B) Polyethylene
C) Polyaniline
D) Polyacrylonitrile
- In an electroless plating of copper, copper will undergo.....reaction.
A) Reduction
B) Oxidation
C) Substitution
D) Neutralization
- Potential of the calomel electrode depends on concentration of
A) Hg
B) Hg_2Cl_2
C) KCl
D) KOH
- The reference electrode, that can be used to determine potential of zinc electrode is
A) Glass electrode
B) Platinum electrode
C) Copper electrode
D) Calomel electrode
- In an Electrochemical sensor, the active component which converts chemical reaction into measurable electrical signal are called as.....
A) Receptor
B) Amplifier
C) Detector
D) Transducer
- The solid electrolyte that is used in the electrochemical gas sensing of SO_x is
A) KOH
B) NaOH
C) K_2SO_4
D) H_2SO_4

12. Dry corrosion is also called as.....corrosion.
 A) Chemical B) Oxidation
 C) Electrochemical D) Wet
13. Chemical formula of rust is
 A) Fe_2O_3 B) Fe_3O_4
 C) FeO D) $\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$
14. Corrosion due to difference in water level is.....corrosion.
 A) Soil B) waterline
 C) pitting D) oxidation
15. An example for one-dimensional nanomaterials is
 A) Nanosheets B) Nanoplates
 C) Nanospheres D) Nanowires
16.is the carbon source used in the synthesis of carbon nanotubes by CVD technique
 A) Nitrogen gas B) Argon gas
 C) Hydrocarbon gas D) Hydrogen gas
17. The orientational behaviour of lyotropic crystals is a function of.....
 A) Concentration B) Solvent
 C) Concentration and solvent D) Neither concentration nor solvent
18. Recycling of e-waste has a direct impact on
 A) Environment B) Economy
 C) Both the environment and economy D) Neither the environment nor the economy
19. Which is the first step of e-waste recycling?
 A) Chemical B) Physical
 C) Biological D) Thermal
20. Mesophase refers to.....
 A) Solid state B) Liquid state
 C) Liquid crystal state D) Gaseous state

PART - B: DESCRIPTIVE ANSWER QUESTIONS**Unit – I**

	Marks	BT*	CO*	PO*
1. a) Describe the construction and working of vanadium-redox flow battery.	6	L*2	1	1
b) Explain the construction, working of $\text{CH}_3\text{OH}-\text{O}_2$ fuel cell.	6	L2	1	1
c) Explain the main components of battery.	4	L2	1	1
2. a) A polymer sample contain 5 molecules having molecular mass of 2000, 4 molecules having a molecular mass of 3000 and 3 molecules having a molecular mass of 4000. Calculate number average and weight average molecular mass.	6	L3	2	2
b) Explain the synthesis and application of following polymers: i) silicon rubber, ii) epoxy resin.	8	L2	2	1
c) Define the following: i) electroless plating, ii) conducting polymer	2	L1	2	1
3. a) Explain the construction, working and applications of Li-ion battery.	8	L2	1	1
b) Describe the synthesis, properties, and application of graphene oxide.	8	L2	2	1

Unit – II

4. a) Explain the construction, working and advantages of calomel electrode.	8	L2	3	1
b) Describe the electrochemical sensor for the determination of dissolved oxygen.	8	L2	2	1
5. a) Explain the galvanized coating and anodization of aluminium.	8	L2	4	1
b) Explain the electrochemical theory of corrosion taking Fe as an example.	8	L2	2	1

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6.	a)	Explain the principle and instrumentation of potentiometry and discuss its application in determining the concentration of iron.	8	L2	4	1
	b)	Explain the working principle of electrochemical and optical sensor.	8	L2	3	1
Unit – III						
7.	a)	Describe the synthesis, and applications of carbon nanotubes.	6	L2	5	1
	b)	Explain the extraction of gold from e-waste.	6	L2	5	1
	c)	What is e-waste? Give any four advantages of recycling.	4	L1	5	1
8.	a)	Define light emitting electrochemical cell. Explain the working principle of light emitting electrochemical cell.	6	L2	5	1
	b)	Explain the nematic, chiral nematic and smectic phases of liquid crystals.	6	L2	5	1
	c)	List the sources of e-waste.	4	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
